

## Title

Metformin-Exercise Interaction in Metabolic Syndrome: Protocol for Investigating Cardiorespiratory and Metabolic Adaptations

## Abstract

Metformin and structured aerobic exercise represent cornerstone interventions in metabolic syndrome management, yet the nature of their potential pharmacological-exercise interaction on cardiovascular fitness development and metabolic substrate flexibility remains incompletely characterized. This protocol describes a randomized, placebo-controlled study examining these interactions. Participants will include metformin-naïve adults with metabolic syndrome, randomized to receive either metformin (1000 mg daily) or matched placebo, combined with identical supervised aerobic exercise training (5 sessions weekly, moderate intensity, 500 kcal per session) over five weeks. Primary outcomes measure peak aerobic capacity and maximal lipid oxidation capability. Secondary measures assess exercise intensity-dependent substrate switching, metabolic efficiency, and subjective exertion perception. An exploratory analysis will examine whether participant age modulates medication-exercise interactions. Findings will clarify optimal pharmacotherapy timing relative to exercise prescription in this population.